

DETAILED PROJECT SPECIFICATION AND RISK MANAGEMENT WORKBOOK

A. PROJECT SPECIFICATION

Functional Requirements Specification (FRS)

Project: Fitness Tracking and Meal Planning Mobile Application

Platform: Android (React Native CLI)

Backend: Firebase (Authentication + Cloud Firestore + FCM)

FR-01 | User Login | The system shall allow registered users to log in using email and password via Firebase Authentication. | High | User is redirected to the dashboard upon successful login. Invalid credentials display an error message.

FR-02 | Fitness Activity Logging | The system shall allow users to create, view, and delete fitness activity logs (type, duration, date). | High | Activity log is saved in Cloud Firestore and displayed in the activity list and daily summary.

FR-03 | Meal Planning | The system shall allow users to create and update meal plans for selected dates (breakfast/lunch/dinner/snacks). | High | Meal plan is saved in Cloud Firestore and correctly loads when the same date is selected.

FR-04 | Nutrition & Macro Tracking | The system shall allow users to add food entries and view daily macro totals (calories, protein, carbs, fats). | Medium | Macro totals update correctly after adding food entries and remain correct after app restart.

FR-05 | Dashboard View | The system shall display a dashboard summarizing fitness activities and nutrition data. |

High | Dashboard displays updated summaries and charts based on saved data.

FR-06 | User Logout | The system shall allow users to securely log out of the application. | Medium | User is redirected to login screen and cannot access dashboard without logging in again

Non-Functional Requirements Catalog (NFR)

NFR-04 (Category): Usability

Description: The app shall allow a first-time user to log one workout entry within 1 minute without training (based on a basic usability test).

Use Case / User Story Artifact

Use Case Name: Log Fitness Activity

Actor: User

Precondition: User is logged in

1. User opens the Fitness Tracking screen.
2. User inputs activity details (type, duration, date).
3. System validates inputs and saves the record to Firestore.
4. System displays the saved activity in the activity list and updates the summary.

Postcondition: Fitness activity is successfully saved and visible to the user.

User Story Format:

As a user, I want to log my workouts/activities so that I can track my fitness progress over time.

Interface / API Specification

API Name: USDA Food Search API

HTTP Method: GET

Base URL: <https://api.nal.usda.gov>

Endpoint: `/fdc/v1/foods/search?query={keyword}&api_key={API_KEY}`

Request: keyword (example: chicken)

Response (sample): `{ "foods": [ { "fdcId": 123, "description": "Chicken breast" } ] }`

Requirements Traceability Matrix (RTM)

Req ID	Design Reference	Test Case	Status
FR-01	UI-01 Login/Signup Screens	TC-01	Not Started
FR-02	UI-03 Fitness Logging Screens	TC-02	Not Started
FR-03	UI-04 Meal Planning Screens	TC-03	Not Started
FR-04	UI-05 Nutrition/Macro Screen	TC-04	Not Started
FR-05	UI-05 Dashboard Screen	TC-05	Not Started
FR-06	UI-06 Settings/Logout	TC-06	Not Started

B. RISK MANAGEMENT

Risks will be identified during planning and development through team discussions and review of technical requirements. Each risk will be analyzed using Probability (1–5) and Impact (1–5) scales. The Risk Score will be computed as Probability × Impact. Risks will be monitored weekly, and mitigation actions will be assigned to responsible team members. Risks with score above 15 will be prioritized for immediate mitigation and contingency planning.

Risk Register

Risk ID	Description	Probability	Impact	Score	Mitigation
R-01	Poor internet/Firebase downtime affects login or data syncing	3	5	15	Add retry + clear error message; test on different networks; schedule buffer time
R-02	Firestore Security Rules misconfiguration causes access issues/data exposure	2	5	10	Peer review rules; test with multiple accounts; apply least privilege
R-03	USDA API rate limit/unavailable causes food search failure	3	3	9	Cache recent results; allow manual food entry; show retry/fallback message
R-04	Android release build/signing issues (APK/AAB/keystore/Gradle) delays submission	3	4	12	Create keystore early, document build steps, generate test release build weekly.
R-05	Feature creep/time constraint delays completion	4	4	16	Define MVP, freeze scope after Week 2, prioritize core modules, weekly progress check.

Risk Visualization Matrix:

Impact \ Probability	1	2	3	4	5
5		R-02	R-01		
4			R-04	R-05	
3			R-03		
2					
1					

GRADING RUBRIC

Criteria	Excellent (5)	Good (3)	Needs Improvement (1)
FRS Completeness	All functional requirements are complete, clear, and testable	Most requirements are clear; minor missing details	Requirements are incomplete, unclear, or not testable
NFR Quality	Non-functional requirements are measurable and realistic	Some NFRs are measurable; others are vague	NFRs are not measurable or missing
RTM Accuracy	Requirements are correctly mapped to design and test cases	Partial mapping; some missing links	No mapping or incorrect mapping
Risk Analysis	Risks have correct probability/impact scoring and strong mitigation plans	Minor scoring issues; mitigation partially defined	Incorrect scoring and weak/missing mitigation